

TABLE V
MATHEMATICAL CHARACTERISTICS OF EACH UC PROBLEM

Problem	Formulation	Type of optimization model	Binary variables included?	Available solver	Case
Non-decomposed dc-UC	Combination of (1), (2), and (9)	MILP ^a	Yes	CPLEX [25]	A
Non-decomposed ac-UC	Combination of (1), (2), and (3)	MINLP ^b	Yes	No off-the-shelf solver	-
Decomposed ac-UC	Benders' algorithm (depicted in Fig. 2) including subproblems (5) and master problem (7)	NLP ^c (subproblems)	No	CONOPT [26]	B and C
		MILP (master problem)	Yes	CPLEX [25]	

^aMixed-integer linear programming

^bMixed-integer non-linear programming

^cNon-linear programming (continuous)

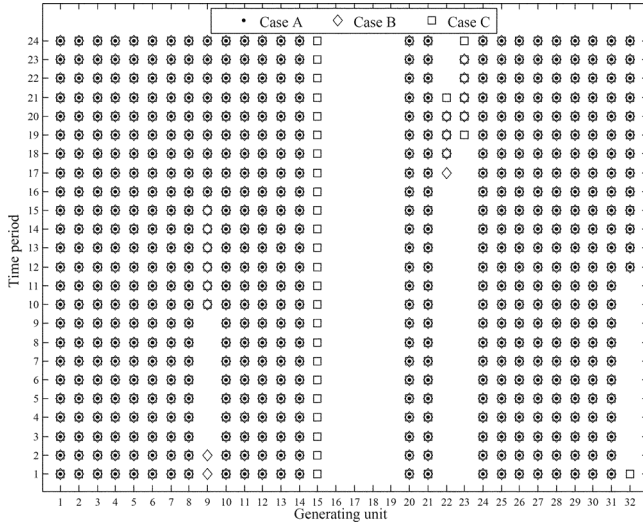


Fig. 4. Commitment status of the units for Cases A-C.

Case B) This case refers to an ac-UC problem based on model (1)–(3) solved by the proposed Benders' algorithm. The voltage magnitude of each node is enforced to be within 0.9 p.u. and 1.1 p.u.

Case C) This case is similar to Case B, but constraints on the voltage magnitude of nodes are relaxed. In this case, the voltage magnitude of nodes can lie within 0.5 p.u. and 1.5 p.u. Although this case is not realistic, it is considered for illustrative purposes.

Table V mathematically characterizes the non-decomposed dc-UC problem (Case A), the non-decomposed ac-UC problem, and the decomposed ac-UC problem (Cases B and C). Note that the non-decomposed dc-UC problem (Case A) includes the dc version of ac constraints (3), which is given in Appendix B as constraints (9).

The corresponding numerical results are given and analyzed in the next subsection.

C. Numerical Results

The expected cost, i.e., the value of objective function (1), for the three cases considered are given in the second column of Table VI. Additionally, the commitment status of the units are depicted in Fig. 4. Note that for all cases, neither load curtailment nor wind power spillage occurs.

TABLE VI
NUMERICAL RESULTS FOR CASES A-C

	Expected cost [\$]	CPU time [minute]
Case A	638537.8	70
Case B	651909.9	194
Case C	650368.9	159

According to Table VI and Fig. 4, the following observations are in order:

- 1) The expected cost in Case A (dc-UC problem) is comparatively smaller than that of Cases B and C (ac-UC problems). The reason of this is that the ac-UC problem enforces more realistic constraints than those of the dc-UC one. Also, the expected cost of Case B is comparatively higher than that of Case C, where voltage magnitude constraints are not tightly enforced.
- 2) The dc-UC and ac-UC problems yield different commitment status for the generating units. According to Fig. 4, generating units 22 and 23 are decommitted in Case A (dc-UC problem), while they are committed in peak hours of Case B (ac-UC problem). Similarly, generating unit 9 is committed in time periods 1 and 2 of Case B.
- 3) In addition to the commitment status of generating units, the dispatch results of committed units from the dc-UC and ac-UC formulations are different. For example in Cases A and B, generating unit 32 is scheduled in peak hour t_{20} 3.2 p.u. and 2.5 p.u., respectively.
- 4) The reason for these differences is the constraints that are not modeled in the dc-UC formulation (Case A), i.e., reactive power constraints (3g) and voltage magnitude constraints (3j). For example, the numerical results of Case B (ac-UC problem) illustrate that the voltage magnitudes of node 6 within different scenarios and time periods are mostly equal to its upper bound, i.e., 1.1 p.u. However, the results obtained by dc-UC model (Case A) misrepresent such constraint, since it does not consider the voltage magnitude bounds. In addition to constraints (3g) and (3j), power losses in the network is the other reason of differences, as it is modeled in the ac-UC problem (Case B), but it is not in the dc-UC problem (Case A).
- 5) According to Fig. 4, although both Cases B and C solve ac-UC problems, different bounds on voltage magnitude